

 	Curso 2023-24 Seguridad de la Información Exercises for Block I DH and modular exponentiation	
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Objective

Prepare for the first test:

- ☐ Fast Modular Exponentiation Algorithm
- ☐ Diffie-Hellman key exchange
- ☐ Attacks on Diffie-Hellman

Software

- ☐ SAMCrypt v1.0: https://www.criptored.es/software/sw_m001t.htm
Note: Pre-installed in ETSISI(Windows) virtual desktop: <https://escritorio.upm.es/>
- ☐ Modular inversion online: <https://dsri.github.io/modinverse/>

Modular Exponentiation.

Exercise 1. Compute by hand $72^{13} \bmod 1.829$. Check the result with SAMCrypt.

Exercise 2. Compute by hand $1.094^{937} \bmod 1.829$. Find how many digits 1.094^{937} has.

Diffie-Hellman

Alice and Bob use Diffie-Hellman's algorithm to perform a key exchange. They choose the prime number 661 with generator $\alpha = 6$.

- a) Alice chooses the private value $a=12$. What message does she send to Bob?
- b) Bob chooses the private value $b=8$. What message does he send to Alice?
- c) Given the above, compute the shared secret key that Alice and Bob can obtain.
- d) Eve can listen to the communication channel. She has unlimited computational power. How could she make a brute-force attack?
- e) Suppose Eve does not want to use brute force. Is there anything else she can do to find out the secret key?
- f) Suppose now that Eve can not only listen, but also intervene in the communication. What kind of attack can she do to hijack the conversation?